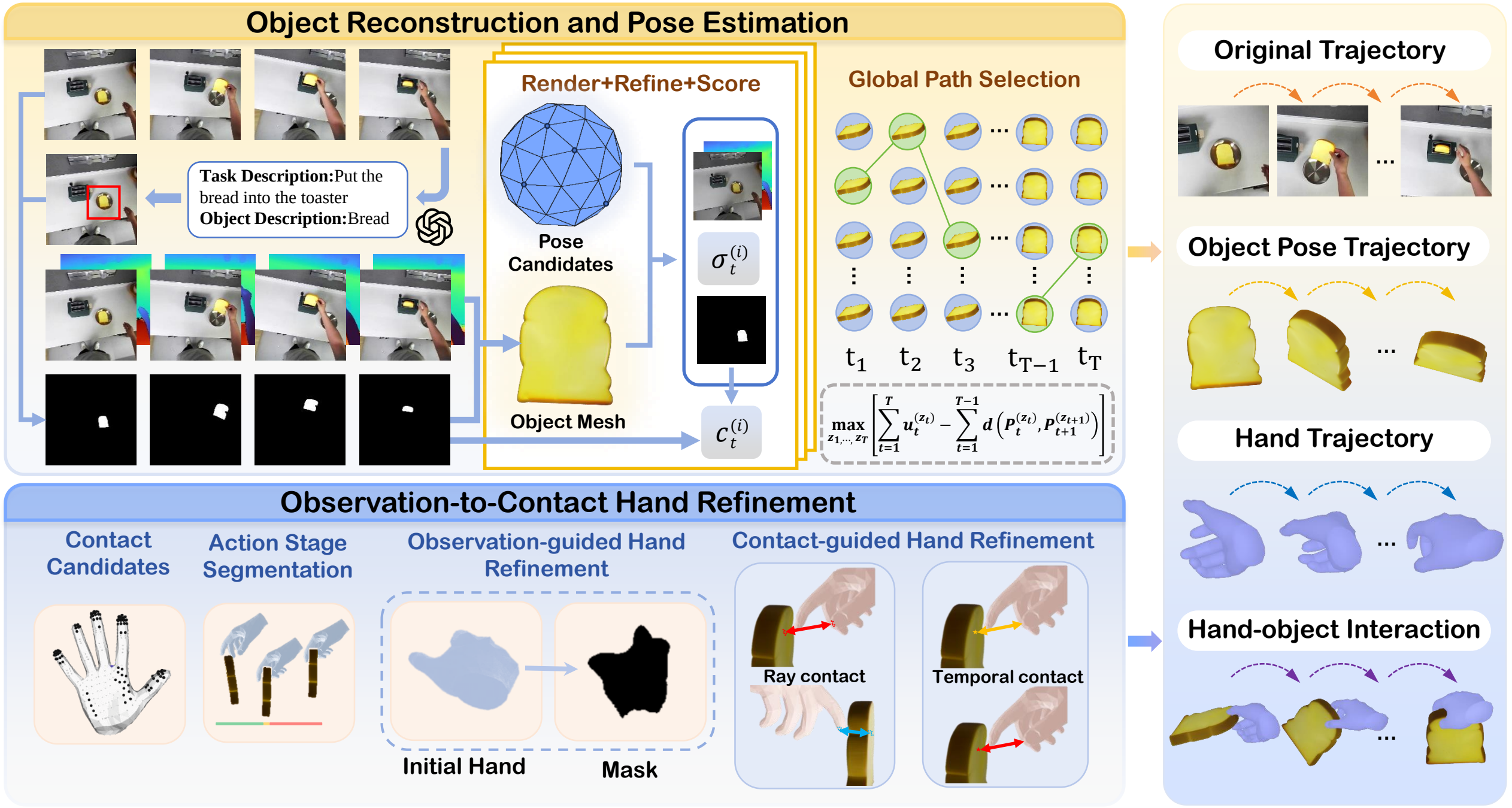

Object Reconstruction and Pose Estimation
Task Description:Put the bread into the toaster
Object Description:Bread
Render+Refine+Score
Pose Candidates
Object Mesh
Global Path Selection
t_1 t_2 t_3 t_{T-1} t_T
$\max_{z_1,\cdots,z_T}\left[\sum_{t=1}^{T} u_t^{(z_t)} - \sum_{t=1}^{T-1} d\left(P_t^{(z_t)}, P_{t+1}^{(z_{t+1})}\right)\right]$
Observation-to-Contact Hand Refinement
Contact Candidates
Action Stage Segmentation
Observation-guided Hand Refinement
Initial Hand
Mask
Contact-guided Hand Refinement
Ray contact
Temporal contact
Original Trajectory
Object Pose Trajectory
Hand Trajectory
Hand-object Interaction